

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP40018
Subject Title	Medical Imaging: Science and Analytics
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Exclusion: AP30017
Objectives	To provide students with an introductory overview on a wide range of common medical imaging technologies, including medical optical imaging, ultrasound, magnetic resonance imaging (MRI), X-ray radiography and computed tomography, and radionuclide imaging. The physical principles and applications of each technology, hardware design, risk and benefits are introduced. Students will also be introduced with the basic of image analytics adopted in the clinical workflow and its latest development.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. describe the principle of optical image formation and design features of various kinds of medical optical imaging devices; b. recognize and discuss the applications and limitations of ultrasound imaging techniques; c. explain the mechanism of magnetic resonance imaging and describe the safety precautions associated with the MRI process; d. describe the principle of x-ray radiography and discuss its associated risks; e. explain the use of radionuclide in various kinds of radionuclide imaging technologies; and f. recognize the basic of image analytics adopted in clinical workflow.
Subject Synopsis/ Indicative Syllabus	Medical optical imaging Principles of optical image formation. Design and applications of scanning laser ophthalmoscopy, and optical coherence tomography. Ultrasound medical imaging Sound wave propagation and interaction in biological tissues. Principle of ultrasound image formation. Mechanism of ultrasound generation; design of ultrasound transducers. Scanning and display formats. Doppler ultrasound and its applications. Benefits and risks of ultrasound imaging. Current trends in commercial instruments. Magnetic resonance imaging Effect of magnetic field on protons in biological tissues; effect of magnetic field

	pulses. Relaxation processes and their measurements. Magnetic resonance imaging. MRI instrumentation, applications and precautions. Current trends in commercial instruments. X-ray radiography and computed tomography Fundamentals of X-rays; interaction of X-rays with the body. Instrumentation for X-ray radiography. X-ray contrast agents. X-ray imaging techniques (planar radiography, mammography and fluoroscopy). Computed tomography: instrumentation and image reconstructions. Applications and precautions for X-rays. Radionuclide imaging Radioactivity and radiotracer half-life. Planar scintigraphy and its applications. Single photon emission computed tomography (SPECT) and its applications. Positron electron tomography (PET): tracers, instrumentation and clinical applications. Image analytics Digital image acquisition. Image enhancement. Feature detection. Segmentation, registration and classification. Basic and latest development of deep learning in medical image analysis.																																						
Teaching/Learning Methodology	Lecture: The concepts related to selected topics will be explained. Examples will be used to illustrate the concepts and ideas in the lectures. The students are free to request help. In-class assignment or quiz will be given to assess the learning progress of students. Tutorial: Various teaching and learning activities will be conducted in tutorial sessions to consolidate the teaching in lectures. Students will work on problem sets in the tutorials, which provide them opportunities to apply the knowledge gained in lectures.																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>1. Continuous assessment</td><td>40</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Examination</td><td>60</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100 %</td><td colspan="6"></td></tr></table> In class assignments and test (assessment method 1) and a final written examination (assessment method 2) all require demonstration of good understanding about (a) to (f). The continuous assessments aim at checking the progress of students’ study throughout the course, assisting them in self-monitoring of fulfilling the learning outcomes. The examination will be used to assess the knowledge acquired by the students; as well as to determine the degree of achieving the learning outcomes.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	1. Continuous assessment	40	✓	✓	✓	✓	✓	✓	2. Examination	60	✓	✓	✓	✓	✓	✓	Total	100 %						
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed (Please tick as appropriate)																																			
		a	b	c	d	e	f																																
1. Continuous assessment	40	✓	✓	✓	✓	✓	✓																																
2. Examination	60	✓	✓	✓	✓	✓	✓																																
Total	100 %																																						

Student Study Effort Expected	Class contact:	
	▪ Lecture & Tutorial	39 Hrs.
	Other student study effort:	
	▪ Self-study	81 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	T. Farncombe, and K. Iniewski, Medical imaging: technology and applications. 2014. Boca Raton, FL: CRC Press, Taylor & Francis Group. S. Webb, and M.A. Flower, Webb's physics of medical imaging. 2012. Boca Raton, Fla.; London: CRC. N.B. Smith, and A. Webb, Introduction to Medical Imaging: Physics, Engineering and Clinical Applications. 2010. Cambridge: Cambridge University Press. K.D. Toennies, Guide to medical image analysis: Methods and algorithms. 2017. London: Springer London.	